

Aeronautical Propulsion

Project 3

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Background

Goal: Simulate a 3-blade suction fan at Mach 0.8 cruising condition at 12 km standard atmospheric conditions

Atmospheric Conditions at 12 km

Pressure: 19.4 kPa

Density: $0.31194 \frac{kg}{m^3}$

Temperature: 216.66 K ($-56.49^\circ C$)

Speed of Sound

$$a = \sqrt{\gamma RT}$$

$$= \sqrt{(1.4)(287 \frac{J}{kg})(216.66 K)}$$
$$= 295.05 m/s$$

Freestream Velocity

$$V_\infty = Ma = (0.8) \left(295.05 \frac{m}{s} \right) = 236.04 \frac{m}{s}$$

In a turboprop engine, thrust is produced by the propeller. A gas turbine uses energy from combustion gases to drive a gearbox connected to the propeller. The propeller then accelerates a large volume of air resulting in lift to be generated. At each blade, the thrust is produced through lift on each rotating airfoil.

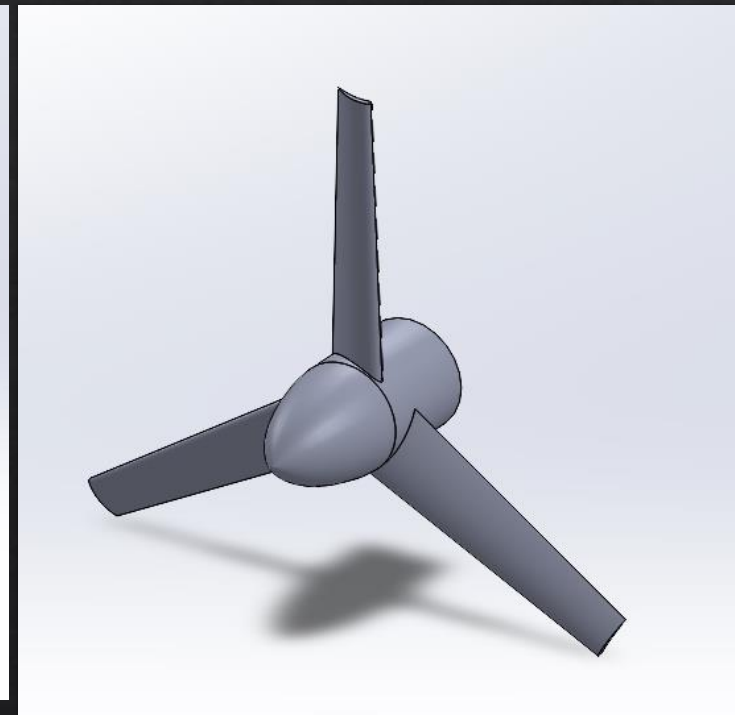
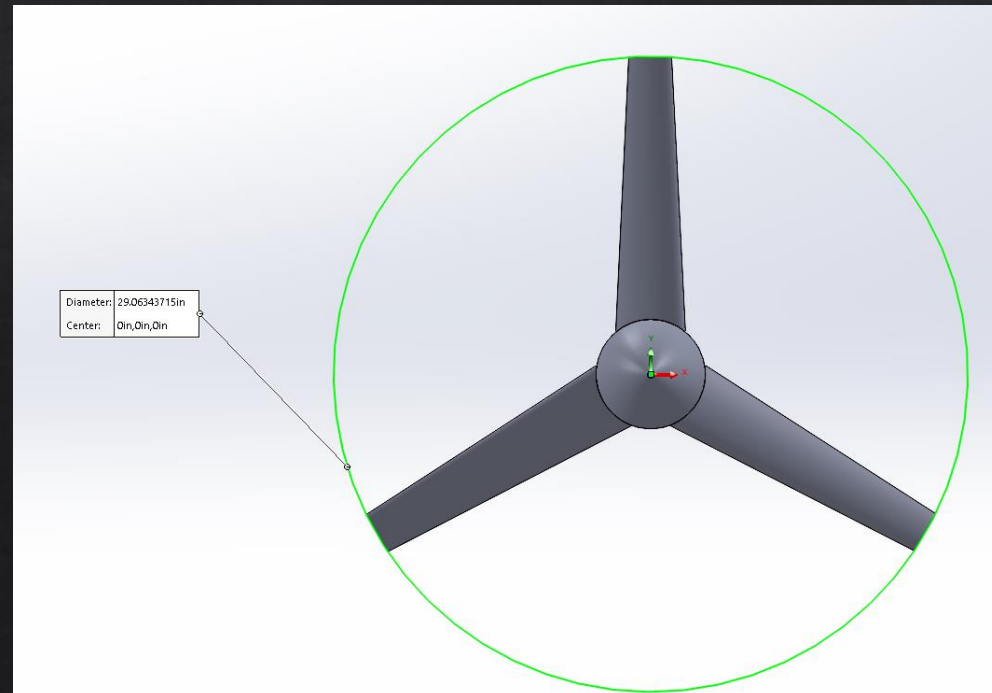
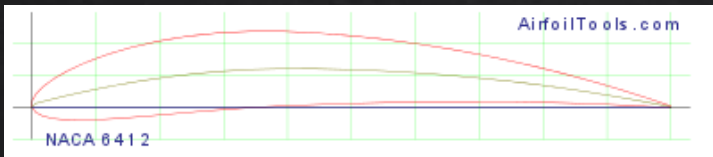
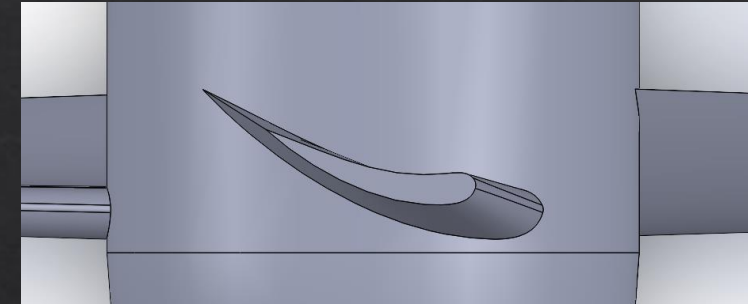
Propellers use Newton's Second Law applied to a control volume. Thrust is generated by changing the momentum of the airflow:

$$F = \dot{m}(V_{exit} - V_\infty)$$

where F is the thrust generated, $\dot{m}$ is the mass flow rate, V_{exit} is the downstream velocity, and V_∞ is the freestream velocity.

Geometry Setup ~ Blades

- 29" tip circle diameter (~31% scale from previous two projects)
- Lofted Airfoil
 - Base: NACA 6412 at 75 degrees from flow direction
 - Tip: NACA 6412 at 85 degrees from flow direction, 50% scaled from base

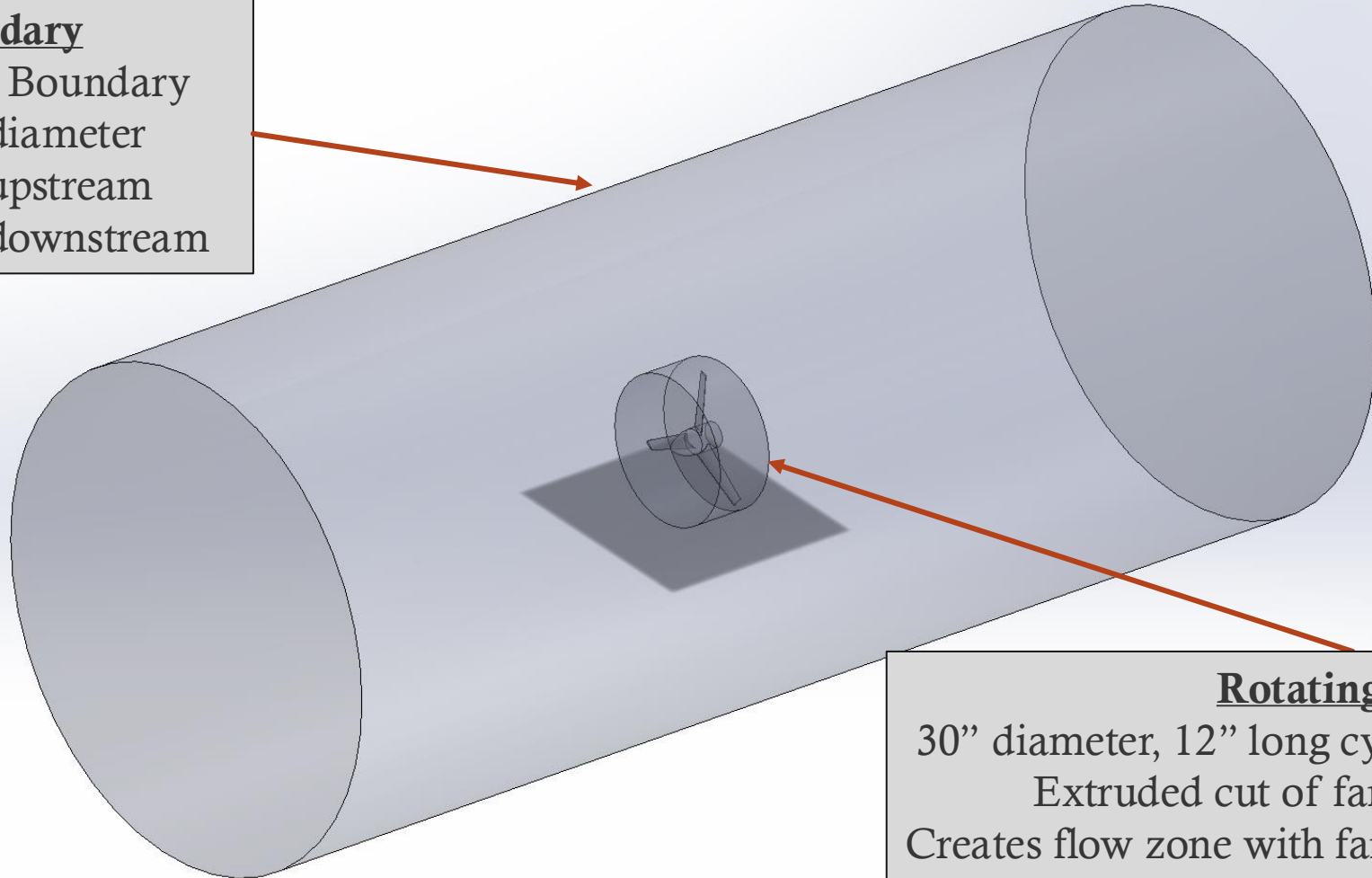


Geometry Setup ~ Rotating Zone and Boundary

Boundary

Cylindrical Boundary

- 100" diameter
- 125" upstream
- 125" downstream

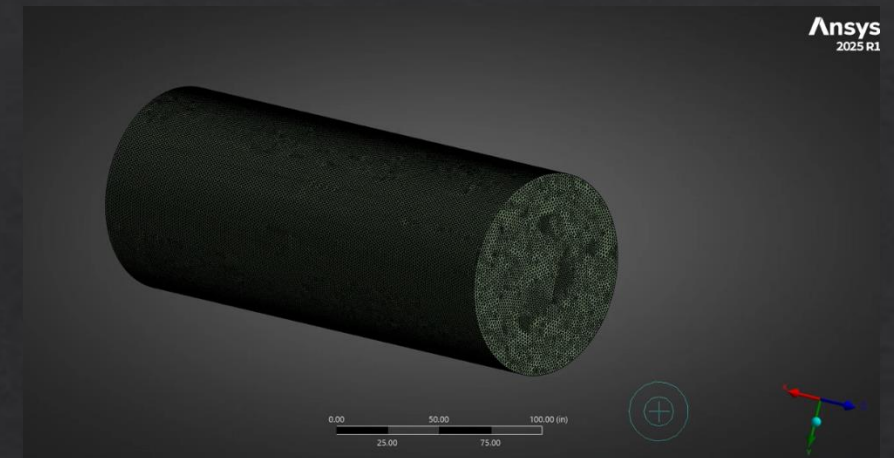


Rotating Zone

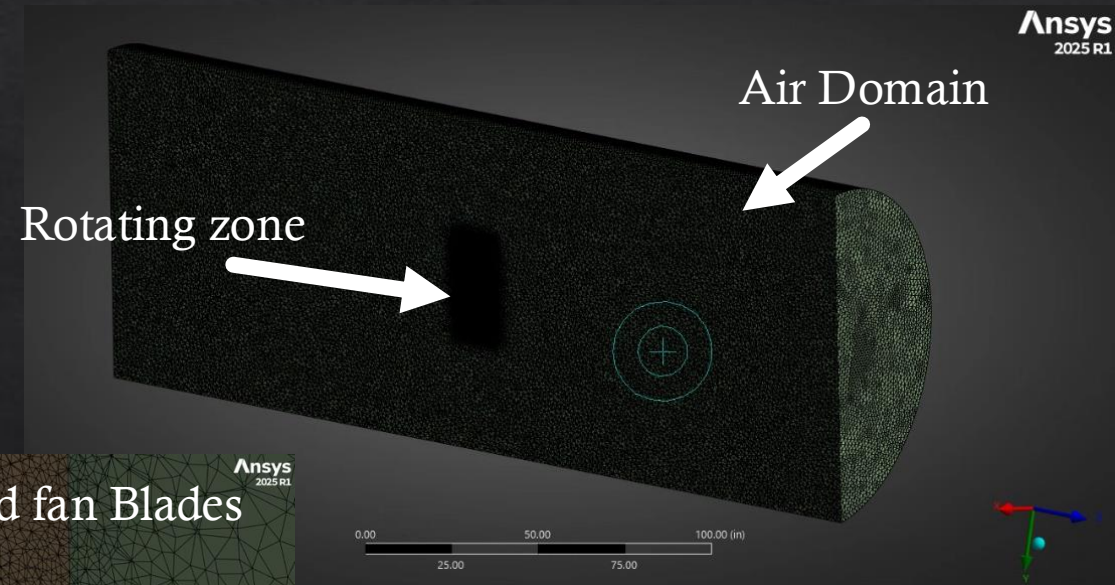
30" diameter, 12" long cylinder surrounding fan
Extruded cut of fan from geometry
Creates flow zone with fan cutout being the walls

Mesh

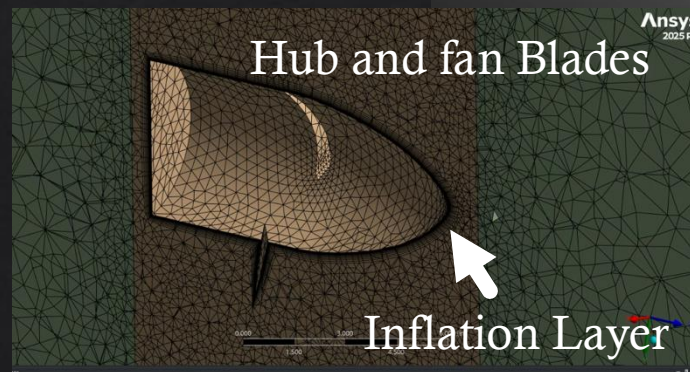
Mesh Control Type	Region	Size	Notes
Body Sizing	Air Domain	Element Size: 0.500"	General Sizing
Body Sizing	Rotating Zone	Element Size: 0.050"	General Sizing
Inflation	Fan Blade Walls	1 st layer size: 0.008" Growth: 1.2 Transition: Smooth	To capture boundary layer effects
Total Number of Elements		6.68 million elements	



Overall Mesh



Cross Section of Mesh



Boundary Conditions and Parameters

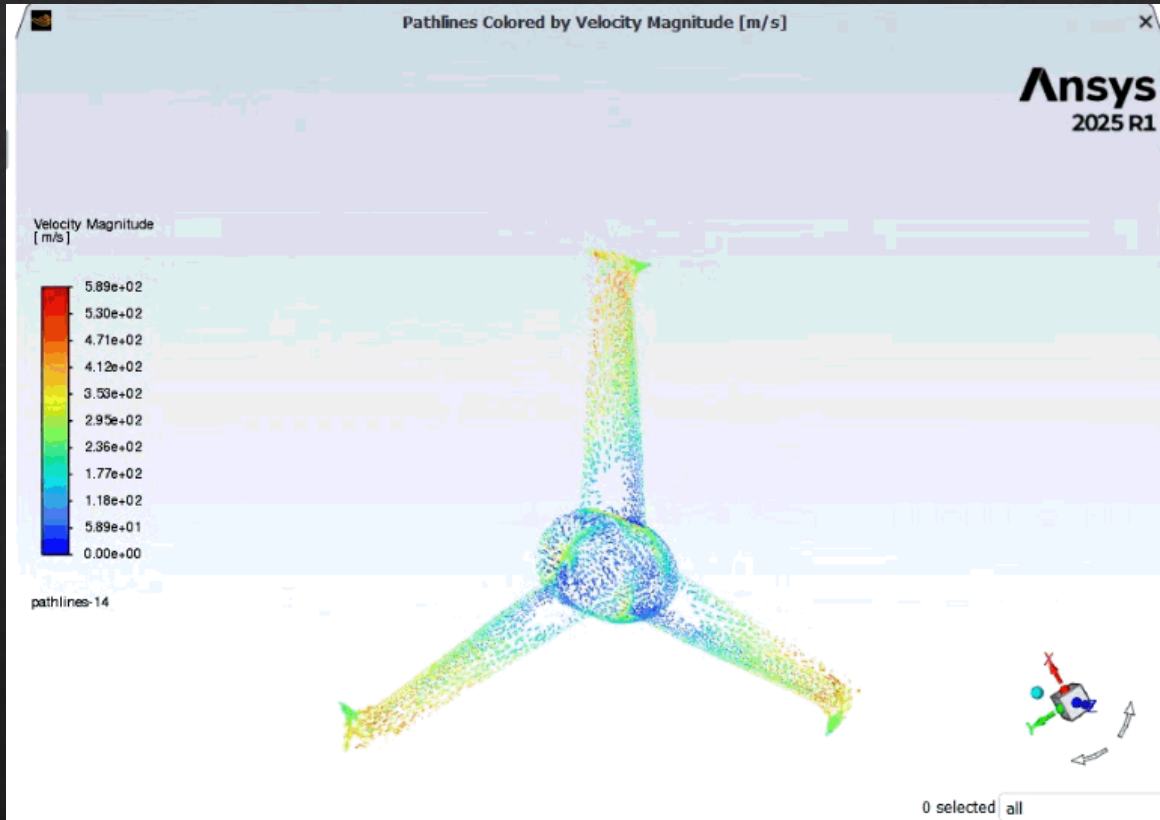
Boundary Conditions

Parameter	Value	Justification
Inlet of Mesh: Velocity Inlet	V: 245 m/s T: 216.65 K P: 19.5 kPa	Simulates Fan moving at Mach 0.8 at 12 km
Outlet of Mesh: Pressure Farfield	T:216.65 K P:19.5 kPa	Simulates ambient atmosphere after fan
Walls: External Boundary Walls Fan Blade Walls	Friction Factor: 0.5 No material selection (Assumes Adiabatic)	Walls to allow for ease of simulation. Assumes no conduction through walls
Rotating Zone: Rotating Flow Region	Frame Motion around stationary wall Speed: 10,000 RPM Direction: Rotating around Z-axis	Simulates fan blades turning in flow
Merged Zones: Merged Walls between Rotating Region and External Boundary	Merged	Must due this so that the flow can travel through both regions as opposed to creating walls

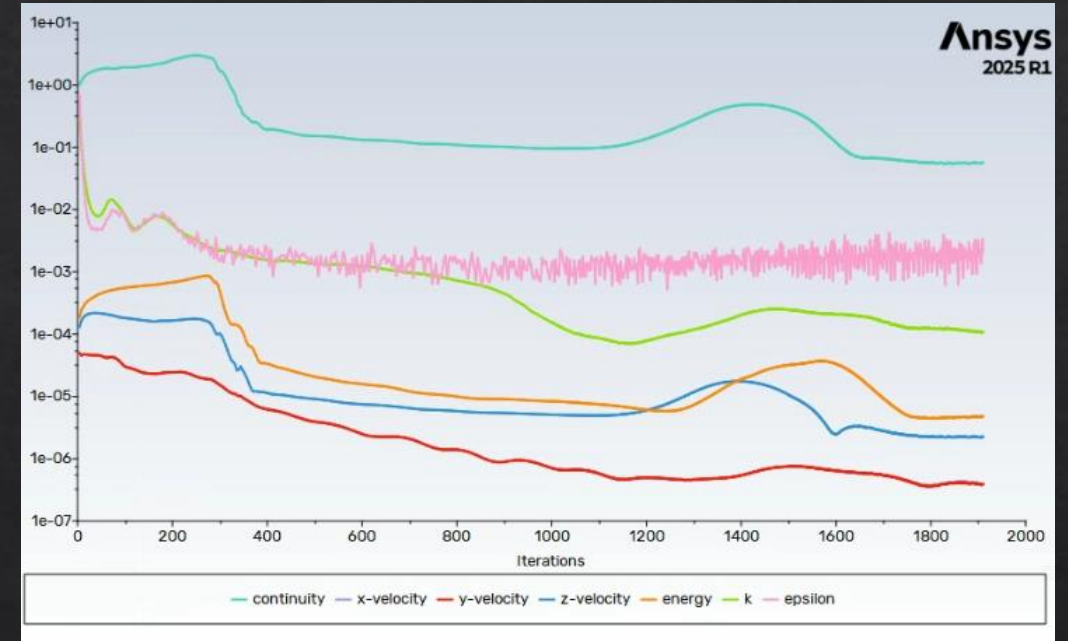
Parameters

Parameter	Value	Justification
Solver	Density-Based	Better for simulating highly compressible flow
Time	Steady	Models steady state and simplifies simulation
Energy	On	Solves energy equation
Viscous	K-ε, Realizable	Better for highly turbulent flow
Fluid, Air	Ideal gas, Sutherland viscosity	Reduces complexity and models temperature dependent viscosity
Methods, Spatial Discretization	Second order Upwind	Greater accuracy, more data points
Initialization	Hybrid	Better for complex flows, fast convergence
Convergence Criteria	Residuals < 1*E-6	Ensures high accuracy

Results



Velocity path lines animation of fast flow turning around fan blades

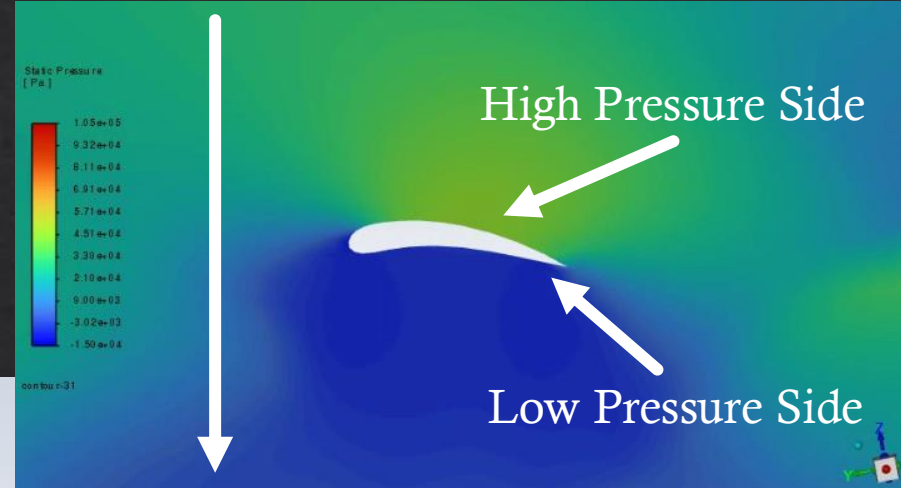


Residuals Plot

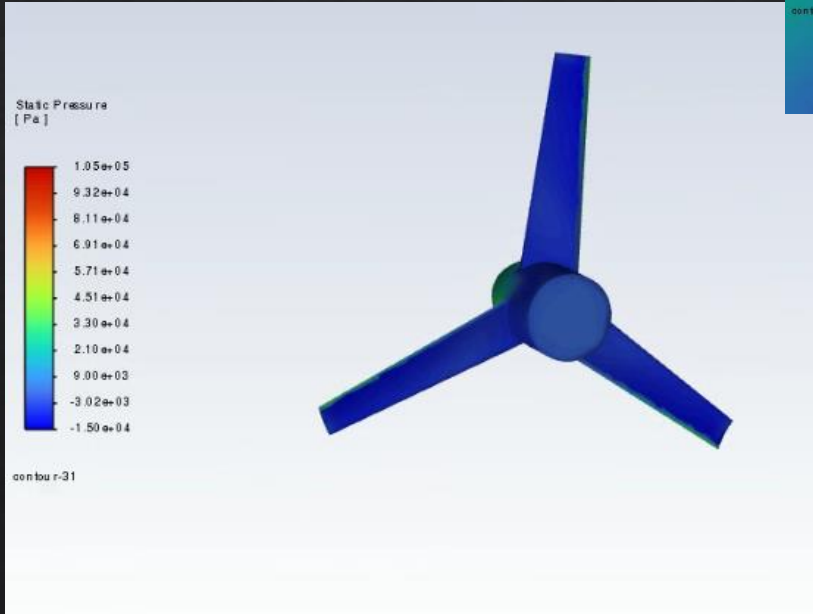
Not all equations were converged, due to the simulation crashing and having to restart and due to the time per iteration there was not enough time to reach full convergence. Please review the report for more information

Pressure Contours

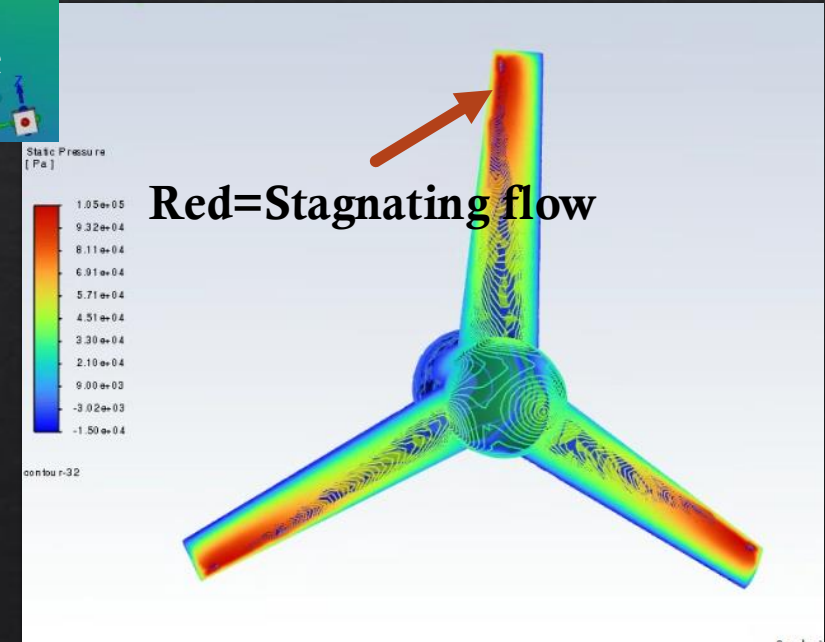
Flow pushed from propeller



Cross Section of Fan Blade Pressure Contour

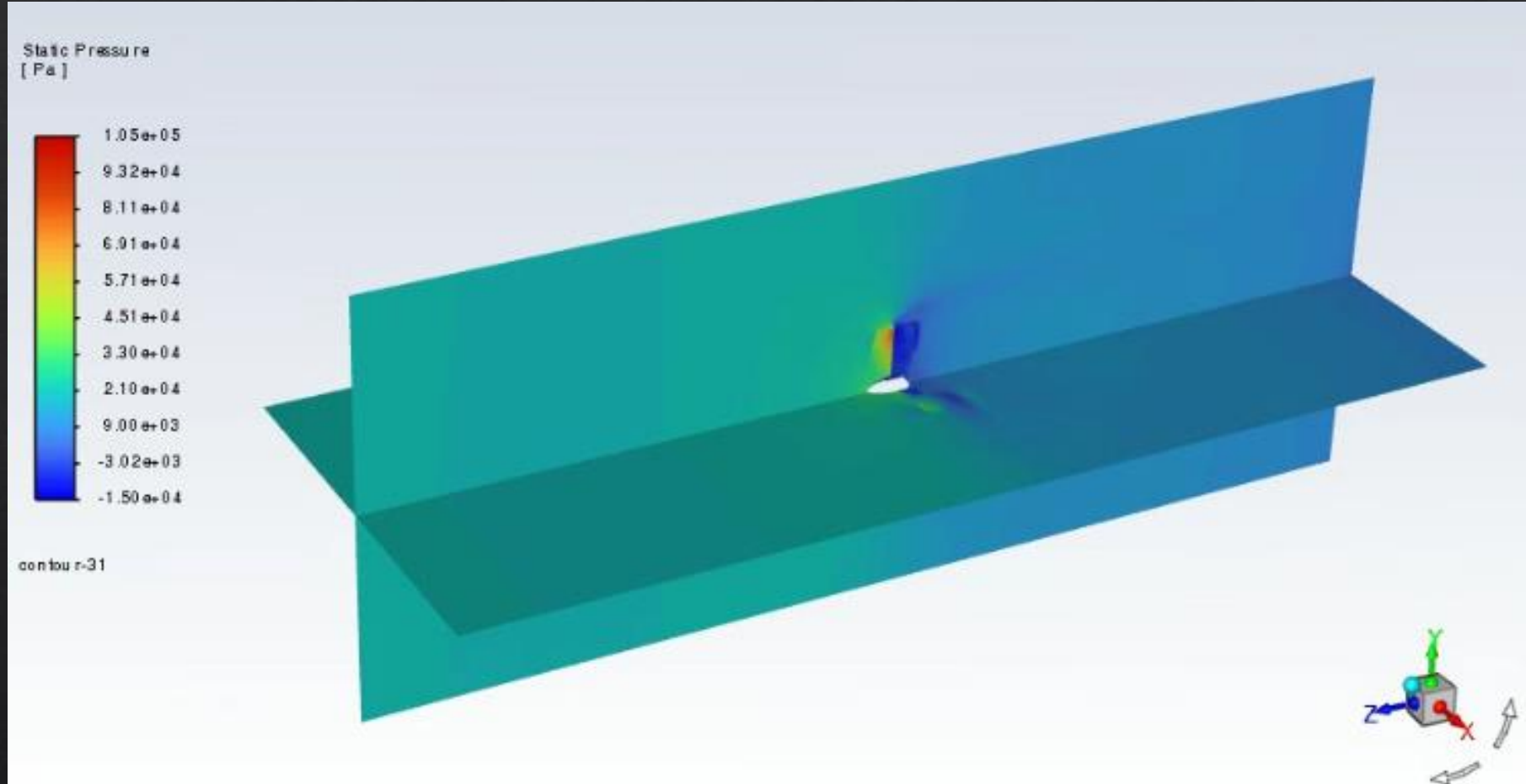


Static Pressure Contour Along Fan Blades (Suction Side)



Static Pressure Contour Along Fan Blades (Pressure Side)

Pressure Contours



Static Pressure Contour of the Entire Domain

Physical Explanation

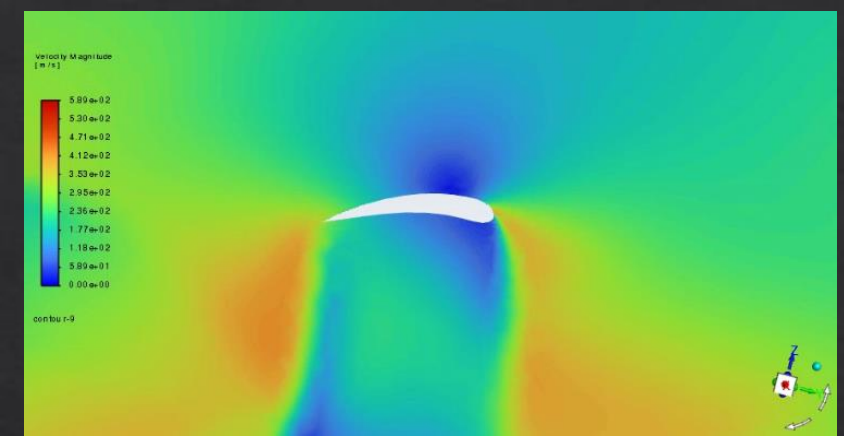
◆ Suction Effect

Occurs when blades move through air at an angle, air is forced to travel faster around one side of the blade than the other. This is caused by the blades angle and curvature. The air that is accelerated creates a decrease in static pressure, and the air that decelerated creates an increase in static pressure. This creates a forward force from this suction effect.

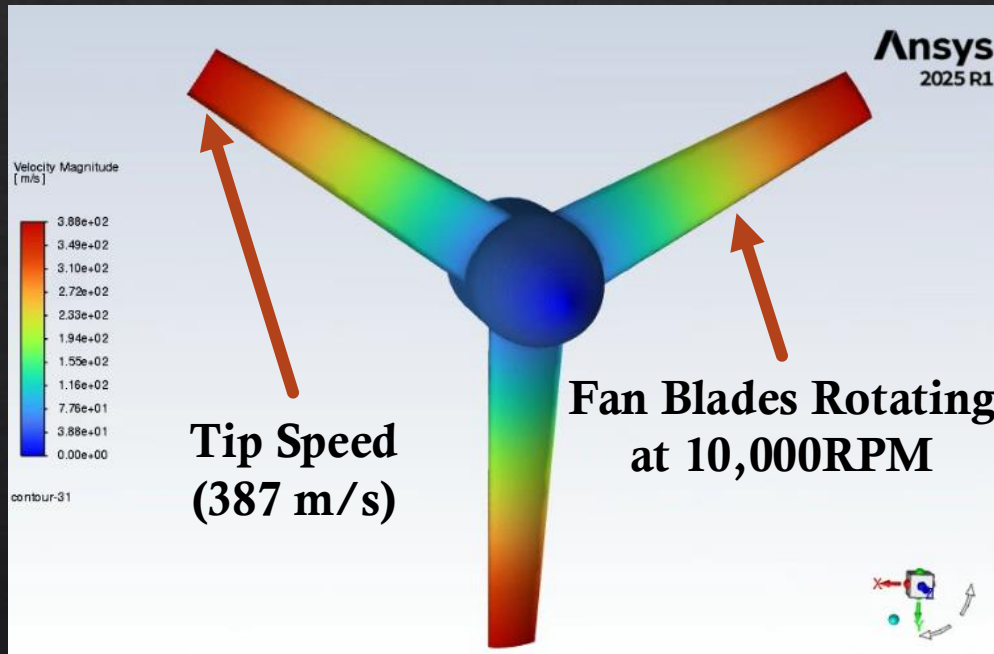
◆ Lift/Thrust from pressure distribution

The airfoil shape of the blades is the main reason for this pressure difference; this creates the lift force. In a rotating blade, this lift force has a forward component and becomes thrust. This is due to the blade's rotation and angle between blade and flow.

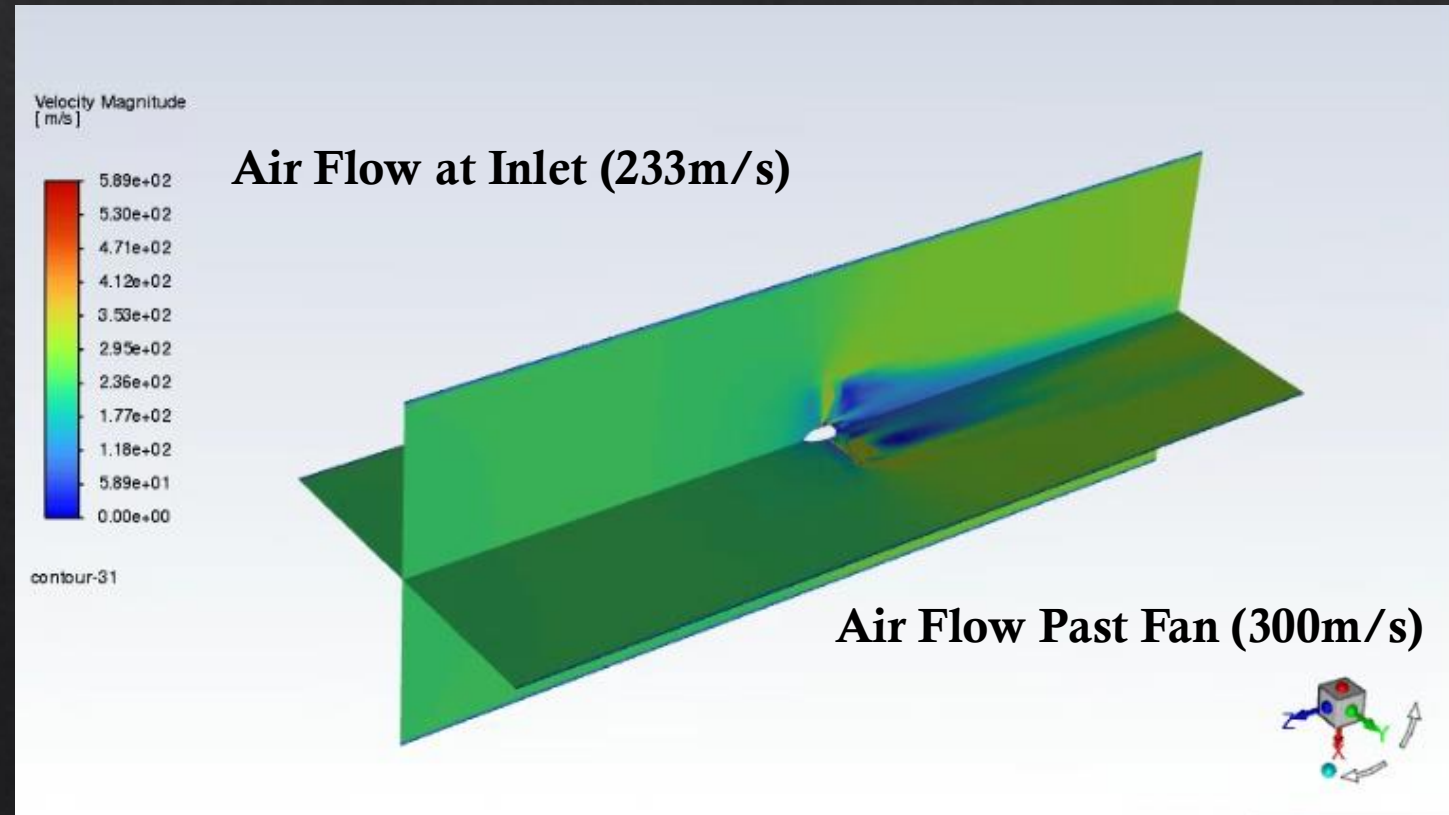
Velocity Contours



Cross Section of Fan Blade Velocity Contour



Fan Blade Velocity Contour



Air Domain Velocity Contour

Physical Explanation

- ◆ Highest velocity at tip of blade

This occurs from a higher tangential speed of the blade at the tip because tangential speed increases with radius. As a result, the airflow experiences its highest relative velocity at the tip.

- ◆ Propellers in aircraft

Main steps in propeller blades summarised:

High rotation of blades --> Flow enters and meets the blades at an angle --> Airfoil shape of blades creates pressure difference --> Lift is generated, which due to blades rotation has a forward component (contributing to thrust) --> Propeller accelerates air backwards (equal and opposite force generating thrust)